8834
H1 Geography Paper 1 Suggested Pointers
TJC Preliminary Examination

1

(a) Explain why this research question is appropriate for the investigation. [3]

- Target group
- Well-defined variables
- Relationship
- Manageability

(b) With reference to Resources 1 and 2, suggest how the students could minimise potential risks when visiting the Tampines neighbourhood for their investigation. [4]

- Mitigate stranger danger
- Watch out! oncoming non-motorised traffic
- Risk from perceived touting
- Sun exposure

(c) Explain three criteria the students could use to select participants for the waste visual assessment and interview. [3]

- Age
- Gender
- Socio-economic profile

(d) Use a suitable data representation technique to represent the tallied data shown in Question 1 in Resource 3. [5]

A suitable technique to use would be the use of pie charts or bar graphs.

(e) Using Resources 3 and 4, answer the research question. [5]

Award one point for each well-explained reason. Award one more mark for each supporting detail.

- Trend seen for the Tampines community making lifestyle changes to reduce their carbon footprint.
- Analyse from looking at:
    - Transport
    - Purchases history
    - Recycling efforts

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(f) With reference to all the resources, evaluate the reliability and accuracy of the investigation. [10]

When thinking about reliability and accuracy of Geographical investigations, consider:

- Reliability refers to the consistency of the data collection methods. If the same procedures are followed each time the investigation is conducted, the results can be considered more reliable. For example, using standardized visual assessment criteria and structured interview questions ensures that all participants are evaluated uniformly, reducing variability in the data.
- If the investigation can be repeated under similar conditions and yield comparable results, this enhances its reliability. The methods should allow other researchers to replicate the study easily. Clear documentation of the procedures, including how visual assessments are conducted and how interviews are structured, is crucial.
- The accuracy of the findings is affected by how representative the sample of participants is. If the sample is biased (e.g., if only certain demographic groups are included), the findings may not accurately reflect the views and behaviors of the entire community. Ensuring diverse participant selection based on demographic representation, occupational relevance, and awareness levels is essential for improving accuracy.
- The tools used for the visual assessment (e.g., checklists, rating scales) and interview questions must be well-designed to accurately capture the necessary information. Validated instruments can increase the accuracy of the data collected. If the questions are ambiguous or leading, they may result in inaccurate or biased responses.
- The techniques used to analyze the collected data influence the accuracy of the findings. If qualitative data from interviews are analyzed using thematic analysis, the coding process should be systematic and transparent to ensure that conclusions are drawn accurately. Similarly, any quantitative data from visual assessments should be statistically analyzed using appropriate methods to ensure validity.
- Researchers must interpret the findings cautiously, considering context and potential limitations. Acknowledging biases and limitations in the study design or data collection process can enhance the credibility of the findings. It is also essential to avoid overgeneralizing the results beyond the studied population.

Marked by level descriptors

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Levels	Marks	Generic Level Descriptors for H1 Fieldwork Evaluative Question
3	9–10	Evaluation is analytical and coherent. Response addresses the question and demonstrates good knowledge and understanding of fieldwork methods relevant to the given context.
2	5–8	Evaluation is mostly analytical and coherent. Weaker responses in this level will have evaluation that is broadly analytical and generally coherent. Response generally addresses the question and demonstrates adequate knowledge and understanding of fieldwork methods relevant to the given context.
1	1–4	Response is descriptive with limited or no analysis and evaluation. Response is fragmented and lacks clarity. Response lacks focus on the question and may be largely irrelevant to it. Response shows basic knowledge and understanding of fieldwork methods. Response has some, though limited, relevance to the given context.
0	0	No creditworthy response

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Section B

Cluster 1 Sustainable Future and Climate Change

Answer both questions

2

(a) Using Resource 5, describe the distribution of cities ranked on the city liveability index. [5]

- High Liveability Concentration in Europe, Canada and Oceania
- Moderate Liveability in cities are mainly concentrated in industrialising macro-regions of Middle East and East Asia
- Low Liveability cities are mainly found in Conflict and Developing Regions:
- Regional Variability in Latin America

(b) Using Resources 6A and 6B, contrast the changes shown in the centre of Birmingham after urban imaging. [5]

Each identified change should relate to a specific sustainability dimension as far as possible, ad substantiated with supporting evidence.

- Economic sustainability
- Social sustainability
- Environmental sustainability

(c) Using an example, explain how there is both complementarity and tension between sustainable urban development and liveability. [3]

Sustainable urban development and liveability are two crucial concepts in urban planning and policy that often intersect but can also conflict. Understanding the complementarity and tension between these concepts is essential for creating cities that are both environmentally sustainable and desirable for residents.

Complementarity
Both sustainable urban development and liveability aim to enhance the quality of life for urban residents. Sustainable urban development focuses on minimizing environmental impact while promoting social equity and economic viability. Liveability, on the other hand, emphasizes creating environments that are safe, accessible, and enjoyable for people. When urban development incorporates sustainability principles—such as green spaces, renewable energy, and efficient public transportation—it can improve liveability by enhancing air quality, reducing noise, and providing recreational opportunities.

Effective urban planning that integrates sustainability into its framework can lead to mutually beneficial outcomes. For instance, creating walkable neighborhoods with mixed-use developments promotes both sustainable transportation options and improved accessibility to services. This not only reduces dependence on cars (a sustainability goal) but also fosters a sense of community and enhances liveability by providing residents with convenient access to shops, parks, and schools.

Sustainable urban development often emphasizes resilience to climate change and environmental challenges. By investing in sustainable infrastructure—such as green roofs,

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permeable pavements, and energy-efficient buildings—cities can improve their adaptability to climate-related events. This resilience contributes to liveability by ensuring that urban areas remain safe and functional during adverse conditions, such as flooding or heatwaves.

Tension
Tensions may arise when the resources allocated for sustainable initiatives compete with those needed for immediate liveability improvements. For instance, funds may be directed toward large-scale renewable energy projects that have long-term benefits but do not address urgent liveability issues, such as affordable housing or public health services. This can create frustration among residents who prioritize immediate quality-of-life concerns over future sustainability goals.

Sustainable urban development initiatives, such as creating eco-friendly neighborhoods or enhancing public transportation, can lead to gentrification. This often results in rising property values and the displacement of low-income residents, creating social inequities. While the development may improve the overall sustainability of an area, it can undermine the liveability of the existing community by displacing long-term residents and altering the social fabric of neighborhoods.

Achieving sustainable urban development requires complex, systemic changes that may conflict with short-term liveability goals. For instance, reducing car dependency through the expansion of public transit may temporarily inconvenience residents who rely on their vehicles for commuting. Similarly, implementing green infrastructure may require construction that disrupts existing urban life, leading to short-term challenges for residents. Balancing these competing demands can create tensions in urban policy and planning.

(d) Using Resource 7, describe the main patterns of resource flows and urban metabolism in Birmingham. [5]

The resource should be interpreted through a systems approach for full credit to be awarded.

Key patterns identified include:
- Birmingham relies on its hinterlands
- Birmingham's material flows are diverse
- Birmingham relies on an extensive transportation network to move resources in and out
- Evidence of circularity, though still automobile dependent

(e) Explain how the demand placed on the environment by urban areas could result in environmental problems in surrounding areas. [4]

① Environmental pressures caused by agricultural and industrial pressures

a. Causing habitat loss and carbon emissions
- Immediate: The dramatic growth of urban areas worldwide leads to the associated encroachment of forested areas. Land uptake through urban sprawl tend to engulf and devour their surrounding lands, often at the expense of biodiversity. For the building of Kalimantan, the new capital city of Indonesia, the deforestation of 100,000 hectares (the estimated area of the new capital) in East Kalimantan has the potential

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to release 48 million tons of carbon into the atmosphere, equivalent to emissions released by 9.3 million vehicles in one year.

- Proximate: Forests that are located next to built cities and suburbs are exposed to high temperatures, pollution, insect pests, invasive plants, and other factors that threaten forest health. Increasing the transport of invasive species, either intentionally or by accident, as people travel and import and export supplies in and out of cities. Urbanization also creates disturbed environments where invasive species often thrive and outcompete native species. For example, many invasive plant species thrive along strips of land next to roads and highways.

b. Air pollution and industrial activity

- The effects of the demand on the environment due to urban developments also affect the broader surrounding areas. *Regions downwind from large industrial complexes* also see increases in the amount of precipitation, air pollution, and the number of days with thunderstorms.

② Environmental pressures caused by resource extraction

- Land degradation is as a result of heavy erosion and loss of land productivity. The land has lost viability for agricultural purposes, as well as loss of habitat for fauna.
- Mining uses heavy machinery and involves blasting during the mineral's extraction important soil organisms have been destroyed, stable soil aggregates disrupted, and eventually depriving the soil of organic matter. These soils are often inhospitable to vegetation.
- Mining operations especially that of the illegal small scale mining are carried out in the open air without appropriate safeguards and environmental standards, and in the process releases contaminated water into the surrounding environment, thus polluting nearby rivers, soils and vegetation.
- Mining activities produce several air pollutants such as nitrogen oxides, sulphur oxides and particulate matter. Gas flares in areas of oil extraction burn constantly, emitting thousands of tonnes of toxic emissions.

③ Environmental pressures caused by landfills (land requirements, seepage water, gaseous emissions)

- Burying organic waste leads to anaerobic decay
- Land subsidence is common as the waste degrades
- Produces methane – a powerful greenhouse gas
- Leachates percolating into groundwater can be toxic

3 Resource 8 shows the average solar radiation and average global surface temperature change, 1910-2010. Resource 9 shows the top emitter countries and their major sources of CO_2 emissions or removal. Resource 10 shows some possible consequences of contemporary climate change.

(a) Using Resource 8, compare the trends in average solar radiation and average global surface temperature change. [4]

Points should be substantiated with data for full credit to be awarded

- Average solar radiation change and average global surface temperature change have rising trends initially.

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- Time lag
- Solar radiation then declines rapidly and then a constant fluctuation whereas temperature change embarks on a continuous rising trend
- Marginal increase for average solar radiation VS substantial increase in average global surface temperature change.

(b) Using Resource 9, describe the CO_2 emissions and the sources by the top emitter countries. [5]

- Fast growing and industrializing countries are the biggest CO2 emitters.
- Developed countries tend to be lower CO2 emitters
- USA is the top ranked emitter country
- Most countries see a varying mix of fossil fuels and land use as their major sources of CO2 emissions.
- The exception is a group of countries with only fossil fuels as their major sources of CO2 emissions.

(c) Explain how the global carbon cycle can be negatively impacted by human activities such as those shown in Resource 9. [4]

Full credit will be given only if the global carbon cycle of storages, flows and disequilibrium is utilised in the answer.

The combustion of fossil fuels (coal, oil, and natural gas) for energy and transportation releases large amounts of carbon dioxide (CO_2) into the atmosphere. This process has significantly increased the atmospheric concentration of CO_2, a potent greenhouse gas. Increase in carbon fluxes from stores to the atmosphere. Elevated CO_2 levels contribute to global warming and climate change by trapping heat in the atmosphere, leading to a range of environmental issues, including more extreme weather patterns and rising sea levels.

Forests act as carbon sinks, absorbing CO_2 from the atmosphere during photosynthesis. Deforestation, whether for agriculture, logging, or urban development, reduces the number of trees as carbon sinks available to absorb CO_2. The reduction in forested areas decreases the Earth's capacity to absorb CO_2, leading to higher atmospheric concentrations of this greenhouse gas. Additionally, when trees are burned or decay, the carbon stored in them is released back into the atmosphere.

Agricultural practices, such as livestock farming, release significant amounts of methane (CH_4), another potent greenhouse gas. Additionally, converting forests or grasslands into agricultural land disrupts the natural carbon storage in these ecosystems. Methane emissions contribute to the greenhouse effect, while land use changes often result in the release of stored carbon in soils and vegetation, further increasing atmospheric CO_2 levels.

The production of cement is a major industrial source of CO_2 emissions. The process involves the heating of limestone (calcium carbonate), which releases CO_2 as a byproduct. Cement production contributes to the buildup of atmospheric CO_2, adding to the overall greenhouse gas emissions associated with human activities.

As atmospheric CO_2 levels rise, more CO_2 is absorbed by the world's oceans. This leads to ocean acidification, a process that alters the chemistry of seawater. Ocean acidification can

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harm marine life, particularly organisms that rely on calcium carbonate to form shells and skeletons, such as corals and shellfish. It also disrupts marine ecosystems and the carbon sequestration capacity of the oceans.

The growth of urban areas and industrial activities increases the demand for energy, which is often met by burning fossil fuels. Urbanization also reduces green spaces, which are important for carbon sequestration. This leads to an increase in CO_2 emissions and reduces the capacity of natural environments to absorb and store carbon, exacerbating the greenhouse effect.

Improper waste management, especially in landfills, results in the decomposition of organic waste in anaerobic conditions, producing methane. Methane from landfills adds to the atmospheric burden of greenhouse gases, contributing to global warming.

(d) Using Resource 10, describe the main patterns of the possible consequences of contemporary climate change. [5]

1. Polar Regions
    - The Arctic and Antarctica are the primary regions experiencing melting ice caps. This is most visible in the northern polar areas and parts of Greenland. These regions show large blue areas, indicating significant ice loss, contributing to global sea level rise.
2. Coastal Regions
    - Coastal risk is prominent along the highly populated areas along East and Gulf Coasts of the United States, Western Europe, South Asia, and Southeast Asia. Red lines along these coastal zones suggest that these areas face the most immediate threats from sea-level rise, flooding, and storm surges due to climate change.
3. Water Scarcity
    - Areas marked by water scarcity are in Southern Africa, the Middle East, parts of India, and Western North America. The blue circles highlight regions that are likely to experience increased droughts and freshwater shortages.
    - These areas also tend to coniincide with areas prone to desertification
4. Desertification
    - Northern Africa, specifically the Sahara region, and areas of the Middle East, Central Asia, Australia, and parts of South America are colored yellow, signifying desertification. These areas are at risk of becoming more arid, with declining soil fertility, reducing the capacity to support agriculture and human settlement.
5. Crop Decline:
    - Agricultural belts in Africa, Asia, and the Americas face growing threats of crop failure and food insecurity due to climate shifts. Regions most affected by crop decline include parts of South America, the Midwestern United States, Sub-Saharan Africa, and South Asia.

(e) Explain how climate change could affect two groups of people differently in countries at lower levels of development. [4]

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Climate change can affect different groups of people in countries at lower levels of development in various ways, often exacerbating existing inequalities. Two such groups that may be affected differently are rural agricultural communities and urban poor populations.

Rural communities in less developed countries often rely heavily on agriculture for their livelihoods. Climate change can lead to more frequent and severe droughts, floods, and other extreme weather events, which directly affect crop yields and livestock. This results in food insecurity, loss of income, and increased poverty for these communities. Rural farmers often lack access to advanced agricultural technologies, irrigation systems, and financial resources to adapt to changing climate conditions. Consequently, they are more vulnerable to climate-induced crop failures and may be forced to migrate or switch livelihoods, which can be extremely challenging due to their limited economic means and social mobility.

In urban areas, the poor often live in informal settlements or slums, which are typically located in vulnerable areas such as floodplains, riverbanks, or coastal zones. These areas are particularly susceptible to climate-related disasters like floods, hurricanes, and rising sea levels. The urban poor may experience more immediate and acute impacts from climate change, such as loss of housing, displacement, and health risks from waterborne diseases due to inadequate sanitation and clean water supply. Unlike rural communities, they may not rely on agriculture, but their ability to respond to and recover from climate disasters is severely limited by their lack of resources, social support, and access to basic services.

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Section C

Answer two questions

4 Evaluate the varying success of strategies to manage non-hazardous solid waste across places.

[13]

he management of non-hazardous solid waste varies significantly across different regions and countries, influenced by factors such as economic development, regulatory frameworks, cultural attitudes, and technological capabilities. When evaluating the varying success of strategies to manage non-hazardous solid waste across different regions, it's important to consider a comprehensive set of criteria that reflect both environmental and social dimensions. Here are some key criteria to consider:

- Percentage of Waste Diverted: The proportion of non-hazardous solid waste successfully diverted from landfills through recycling, composting, and waste-to-energy processes.
- Reduction in Landfill Volume: Measurement of how much waste management strategies reduce the overall volume of waste sent to landfills.
- Cost-Benefit Analysis: Assessment of the costs associated with implementing each waste management strategy versus the economic benefits derived from resource recovery, energy production, and reduced landfill costs.
- Funding and Investment: Availability of funding and investment in infrastructure for waste management systems.
- Reduction in Greenhouse Gas Emissions: Evaluation of how each strategy contributes to lowering emissions related to waste decomposition or processing.
- Soil and Water Quality Impact: Assessment of potential leachate and contamination issues associated with landfills versus the benefits of composting and recycling.
- Participation Rates: The level of public engagement in recycling, composting, and other waste management programs.
- Public Awareness and Education: Effectiveness of outreach and education efforts to inform residents about waste management practices and the importance of participation.
- Availability of Facilities: Accessibility of recycling centers, composting facilities, and waste-to-energy plants in relation to the population.

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- Operational Efficiency: The efficiency of waste collection, sorting, and processing operations.
- Legislation and Standards: Existence of supportive laws and regulations that encourage sustainable waste management practices.
- Incentives for Recycling and Waste Reduction: Availability of financial or tax incentives to promote recycling and other waste management strategies.
- Equity in Access to Services: Ensuring that all communities, regardless of socioeconomic status, have equal access to waste management services.
- Impact on Vulnerable Populations: Assessment of how waste management strategies affect marginalized groups, including potential displacement or environmental injustices.
- Integration of New Technologies: The use of innovative technologies to improve waste management processes (e.g., smart waste bins, advanced recycling techniques).
- Flexibility of Programs: The ability of waste management systems to adapt to changing waste streams and community needs.
- Resource Recovery Rates: The effectiveness of strategies in recovering materials and resources for reuse, and their contribution to a circular economy.
- Environmental Sustainability: Assessment of the long-term environmental impacts of each waste management strategy, including biodiversity and ecosystem health.
- Quality of Data: Availability and reliability of data on waste generation, processing, and outcomes.
- Monitoring and Reporting Systems: Existence of systems for tracking the performance of waste management strategies over time.

5 Evaluate the success of strategies to address the issues faced by the elderly living in cities.

[13]

When evaluating the success of strategies aimed at addressing the issues faced by the elderly living in cities, a range of social, economic, and environmental criteria should be considered. These criteria can help determine the effectiveness of policies and programs in improving the quality of life, accessibility, and well-being of elderly urban residents. Below are key criteria to consider:

- Healthcare Access: Availability of and ease of access to healthcare services, including hospitals, clinics, and specialized care for the elderly.
- Public Transportation: Evaluation of how accessible and elder-friendly public transportation systems are, including factors like low-entry buses, proximity to stops, and frequency of service.
- Proximity to Amenities: The ease with which the elderly can access essential services, such as grocery stores, pharmacies, and community centers, within walkable distances.
- Availability of Affordable Housing: The supply of housing options that are affordable for elderly residents, particularly those on fixed incomes or pensions.
- Elder-Friendly Housing Design: Assessment of whether housing options are designed with age-related needs in mind, including accessible layouts, elevators, handrails, and other safety features.
- Opportunities for Social Engagement: Availability of community programs, clubs, or activities that promote social interaction and prevent isolation among the elderly.
- Support Networks: Presence of support services such as home care, counseling, and social workers who can assist elderly residents in navigating city life.
- Intergenerational Integration: Evaluation of strategies that promote interaction and mutual support between different age groups, fostering a sense of community.
- Crime and Safety Perception: The level of personal safety perceived by elderly residents, as well as actual crime rates in neighborhoods with a high proportion of elderly people.

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- Emergency Response Services: The effectiveness and responsiveness of emergency services, including healthcare, police, and fire departments, in meeting the specific needs of elderly residents.
- Pedestrian-Friendly Streets: The safety and usability of sidewalks, pedestrian crossings, and public spaces for elderly people, including provisions such as benches, ramps, and traffic calming measures.
- Barrier-Free Access: The extent to which public infrastructure (parks, government buildings, libraries) is accessible to individuals with reduced mobility, through the use of ramps, elevators, and designated pathways.
- Mental Health Services: Availability of mental health services tailored to the elderly, such as counseling and support for dealing with loneliness, depression, and dementia.
- Physical Health and Exercise Opportunities: Accessibility to recreational facilities, parks, and exercise programs that encourage physical activity suitable for elderly individuals.
- Pension and Income Support: Adequacy of financial support systems such as pensions, social security, or other financial aid programs to meet the living expenses of elderly residents.
- Affordability of Essential Services: Evaluation of how affordable essential services like utilities, transportation, and healthcare are for elderly residents, especially those living on limited incomes.
- Implementation of Age-Friendly Policies: Existence and enforcement of policies that explicitly address the needs of elderly residents, such as zoning laws, subsidies for senior housing, and healthcare provisions.
- Participation in Decision-Making: Opportunities for elderly residents to have a voice in city planning, policy-making, and local governance.
- Digital Literacy and Access: Availability of programs that help elderly residents become digitally literate, along with access to affordable internet services and devices.
- Elder-Friendly Technology Solutions: Assessment of how well smart city technologies (such as telemedicine, safety monitoring apps, and smart public transportation systems) are adapted to the needs of the elderly.
- Life Satisfaction: Surveys or studies that measure the overall happiness and life satisfaction of elderly residents, including their feelings of belonging and well-being.
- Independence and Autonomy: The ability of elderly residents to live independently with dignity, including access to personal assistance or technologies that support independent living.

6 ‘The uncertainties over future impacts of climate change are the most significant challenge in implementing strategies to respond to contemporary climate change.’

Evaluate this statement.

[13]

The statement that "the uncertainties over future impacts of climate change are the most significant challenge in implementing strategies to respond to contemporary climate change" presents a critical perspective on climate action. To evaluate this statement, it is essential to analyze the nature of these uncertainties, their implications for climate change strategies, and consider other significant challenges that also impact climate response efforts.

- Complexity of Climate Systems: Climate systems are influenced by numerous variables, including greenhouse gas emissions, natural climate variability, and human activities. This complexity can make precise predictions challenging.
- Regional Variability: Climate change impacts can vary significantly across different regions and ecosystems, making it difficult to apply one-size-fits-all solutions.

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- Feedback Mechanisms: Climate change includes feedback loops (e.g., melting ice reducing albedo, releasing methane from thawing permafrost) that can amplify or mitigate impacts in unpredictable ways.
- Risk Assessment: Uncertainty complicates risk assessment and management strategies, as stakeholders may struggle to quantify potential impacts and prioritize actions effectively.
- Adaptation Strategies: Uncertainty can hinder the development of adaptive strategies, as communities may be unsure about which vulnerabilities to prioritize or how to allocate resources effectively.
- Uncertainty can lead to paralysis in decision-making, where policymakers and organizations may hesitate to act due to fear of unintended consequences or ineffective measures. This can result in missed opportunities to mitigate or adapt to climate change.
- Uncertainty about climate change impacts can also affect public perception and engagement. When people are unsure about the consequences of climate change, they may be less likely to support or participate in climate action initiatives.
- Lack of Consensus: Political polarization and lack of consensus on climate change can impede the implementation of effective policies. This is evident in various countries where climate action is subject to changing political administrations and ideologies.
- Short-term Focus: Policymakers often prioritize short-term economic interests over long-term sustainability, making it difficult to commit to necessary but potentially costly climate strategies.
- Funding and Resources: Implementing effective climate change strategies requires substantial financial investment. In many regions, especially in developing countries, limited resources can hinder climate action.
- Transition Costs: The transition from fossil fuels to renewable energy sources involves costs that may be politically contentious and economically challenging, particularly for industries reliant on fossil fuels.
- Disproportionate Impacts: Vulnerable populations often bear the brunt of climate change impacts while having limited capacity to respond. Addressing these social inequalities is essential for effective climate action but can complicate implementation.
- Community Engagement: Ensuring that local communities are involved in decision-making processes can be challenging, especially in areas with marginalized populations or where public awareness is low.
- Innovation Gaps: The development and deployment of new technologies necessary for climate mitigation and adaptation (e.g., carbon capture, renewable energy technologies) can be slow and uncertain.
- Infrastructure Resilience: Many cities and regions lack the infrastructure to withstand climate impacts, and retrofitting or building resilient systems requires substantial investment and planning.

All essays are marked by level descriptors

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Levels	Marks	Generic Level Descriptors for H1 Essays
4	11–13	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	8–10	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	5–7	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–4	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response

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